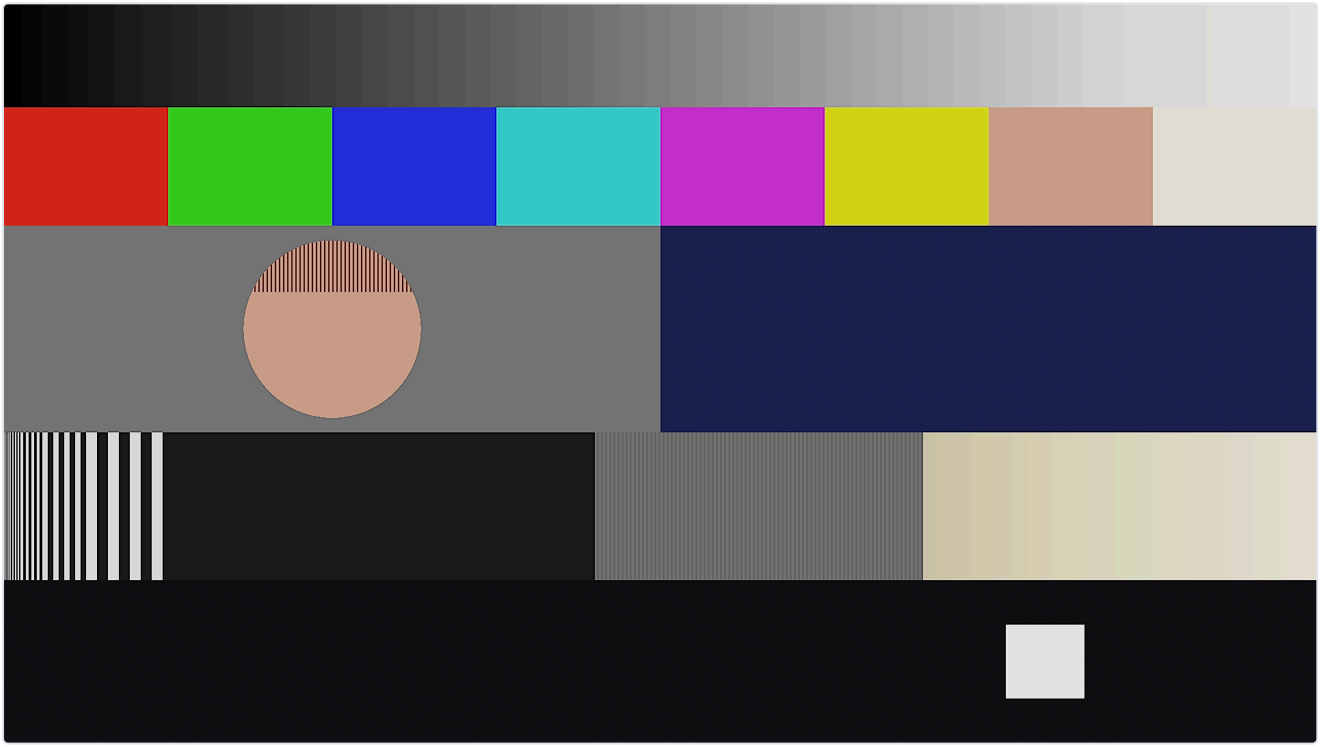
CCU user guide

CCU is a broadcast camera's processing chain with every knob out, for Resolume Arena and Avenue as an FFGL effect, and from the same source as an OpenFX effect for DaVinci Resolve, Vegas, Nuke and Natron (see OpenFX). It runs your clip through the stages of a studio or OB camera in their fixed order, in linear light, with each knob labelled the way a camera control unit labels it.

The stages are master gain, white balance, matrix, detail, knee, gamma and black gamma, pedestal and white clip, and each one is a known circuit. Nothing paints a "video look" on. The badly set-up camera looks fall out of the order: halos the width of the detail delay, grain sharpened by a low coring, faces softened while the jacket stays sharp, highlights gone milky through the knee, a warm white that clips in one channel first, and a white balance that drifts as the camera warms up. None of it is drawn, and all of it will look familiar to anyone who's sat at a CCU panel.



The repo's test card through the plugin at the defaults, rendered by the offline harness rather than captured from Resolume. The halos on the bars are the width of the detail delay, not of the edge. The top of the ramp has gone through the knee, the hair on the skin disc is softer than the weave on the jacket because of the skin window, and the coring has left the floor's fine texture alone.

> **Before you rely on this:** it's released at **v0.2.0**, which adds the OpenFX build, and it's still early days. The chain is measured rather than taken on trust, by a harness that drives the real plugin class and reads each claim back out of the picture it made, at two rasters:
>
> - every stage at its null returns the input within a bound derived from the OETF round trip in float, alpha bitwise, and again after a resize
> - a step's overshoot is Detail Level × h / 4 for exactly the delay, whole and fractional, horizontal and vertical
> - an edge below Coring gets no detail at all, and one above it gets the overshoot less the dead zone
> - above the knee point a ramp's slope is Knee Slope, continuous at the point, per channel
> - a ramp follows the OETF at three exponents, and Black Gamma lifts only below its level
> - red at +6 dB through white balance and then the knee lands where that order predicts, and 34,605 tolerances from the other order
> - the detail gain inside the skin window is 1 − Skin Detail, and outside it's 1
>
> Eight chains broken on purpose each fail their check, and one character changed in the shipped shader is caught. All 23 controls are shown to change the picture.
>
> It has **never been loaded into Resolume on macOS**. The one host it's run in there is the fleet's own test host, `oxbow`, for 120 frames. On Windows, a build of this source loads, registers and renders in Resolume Arena 7.27.1 on software rendering (win-lab, Mesa llvmpipe, no GPU): all 29 host controls match the declaration and all 24 that take a value move the picture, which is 9 of the fleet gate's 9 checks. Software rendering says nothing about a GPU or about speed.
>
> The OpenFX build renders the same pictures as the Resolume build, byte for byte, in the fleet's OFX test host, and in DaVinci Resolve Studio 21.1 on macOS it renders as a Fusion tool and matches that test host to within 1/255. It has **never been run in Vegas, Nuke or Natron**, and its Windows and Linux builds have never rendered in a host. Try it on a spare layer before you put it in a show.
>
> This codebase was created with AI assistance, directed and reviewed by a human author.

Installing

Every download carries one effect, **SW CCU**. Drop it into Resolume's effects folder and restart Resolume:

```
macOS    ~/Documents/Resolume Arena/Extra Effects/
Windows  %USERPROFILE%\Documents\Resolume Arena\Extra Effects\
```

Avenue uses the same layout under its own folder name. The effect then appears in the effects browser as **SW CCU**.

The macOS download is a universal build (Apple silicon and Intel), as a `.dmg` or a `.zip`. The Windows download is an x64 installer or a `.zip`. It isn't code-signed, so the installer trips SmartScreen once: **More info** → **Run anyway**.

Installing the OpenFX build

The OpenFX build is a separate download, `ccu-ofx-<platform>.zip`, for macOS (universal), Windows (x64) and Linux (x86_64). It ships from v0.2.0 on. Copy `CCU.ofx.bundle` from the zip into the standard OpenFX folder and restart the host:

```
macOS    /Library/OFX/Plugins/
Windows  C:\Program Files\Common Files\OFX\Plugins\
Linux    /usr/OFX/Plugins/
```

The effect appears as **CCU** in the **Stoatworks** group. It has the same controls, in the same groups, with the same ranges and defaults as the Resolume build, and everything in this guide applies to it, with the differences listed under OpenFX below.

The one idea

The "video" look of a studio or OB camera comes from a processing chain in a fixed order, and each stage is a circuit a camera engineer would recognise:

```
linear light → master gain → white balance → matrix → detail → knee → gamma and black gamma
→ pedestal → white clip
```

CCU puts the stages in that order, in linear light, and labels the knobs as a CCU does. Then the looks fall out:

- **Halos.** Detail (aperture correction) adds a high-passed copy of the picture made from pixel and line delays, so a hard edge overshoots on one side and undershoots on the other for the width of the delay, not of the edge. A step of height h overshoots by Detail Level × h / 4.
- **Coring.** Detail below a threshold isn't boosted at all, so a low coring sharpens the grain and a high one plastic-coats the picture, leaving only the big edges with their halos.
- **Skin detail.** Inside a hue window the detail gain is reduced, so faces go soft while the jacket stays sharp. Set it wrong and it's the other way round: a red jacket goes soft while the face stays sharp.
- **Knee.** Above the knee point the highlights are compressed instead of clipped, so a sky keeps its colour and a white background goes milky. With the knee off they clip, and because the knee follows white balance, a warm white clips in red first.
- **Black gamma** lifts the shadows on a curve below a quarter of video level and nowhere else.

- **Drifting white balance.** The R and B gains wander slowly on a random walk in mireds, the way they did on a camera that was still warming up: visible over a minute, never a jump.

The clip stands in for scene light: it's linearised through the inverse of the BT.709 curve and its white is the top of what the chain ever sees. A real camera's knee has a scene two or three stops brighter than white to work on. Here **Master Gain and the R and B gains are what push anything above white**, and the knee's whole character on footage comes from them.

Start here

Put SW CCU on a clip or a layer with footage that has bright edges, some highlights and, ideally, a face. At the defaults it's a hot broadcast camera rather than a broken one: Detail Level 0.9 at a 2-pixel delay, coring at an edge of 0.0225, the knee on at 0.7 linear with a 0.29 slope (so white lands at 90% video and the top of the picture goes milky), Black Gamma 0.2, the Standard matrix, a 9-mired drift. Bright edges carry a halo of the delay's width, hot spots compress to a milky grey, and the coring leaves the fine dark texture alone.

Then:

1. **Show Detail on.** The picture is replaced by the detail signal about mid grey: the edges the chain is boosting, and nothing else. Move **Crispening Freq** and watch the lines thicken; move **Coring** up and watch the fine texture vanish from the view. This is how a camera engineer sets detail, and it's the quickest way to understand the first three controls. Turn it off.
2. **Detail Level up to 1.** Every bright edge carries a hard halo. **Crispening Freq up** and the halo widens without the edge moving.
3. **Coring to 0.** The grain and every fine texture is sharpened with the edges. **Coring to 1** and the picture is plastic: only the biggest edges still have detail.
4. **Knee On off.** The milky highlights turn to clipped white. **Master Gain up** with the knee off and the highlights clip hard; knee back on and they compress instead.
5. **R Gain up, knee off.** The whites go warm and the highlights clip in red first: each highlight's core goes white inside a warm surround, because red reaches the ceiling before green and blue do. Knee on and the red is compressed rather than clipped, so the highlight stays warm all the way up.
6. **Gamma** to the bottom for a flat, crushed-highlight picture; to the top for a bright, contrasty one. **Black Gamma** up to lift the shadows without touching anything above a quarter level.
7. **Matrix** through its four presets, and **Saturation** about the middle.

Every slider is declared to the host as 0 to 1. The value each position stands for is given with each control below, and every null (0 dB, a gain of 1, a saturation of 1) is exact.

The Exposure group

Master Gain — head-end gain, in **linear light, before white balance**: **−6 + 24 v dB**, from −6 dB at 0 to +18 dB at 1, exactly **0 dB at 0.25**, the default. A CCU's dB gain is sensor gain, and a gained-up highlight is what the knee exists to catch, so it sits at the front of the chain rather than after gamma as a video gain would. It's the control that pushes a clip's white above 1.0 and hands the knee something to compress.

Slider	gain	what it does at the defaults
0	−6 dB	the picture at half, nothing near the knee
0.25	0 dB	the default
0.5	+6 dB	whites at 2× linear: everything bright is in the knee
0.75	+12 dB	mid-tones in the knee, a milky picture
1	+18 dB	8× linear

Master Black — the pedestal, added on video after gamma: **0.2 (v − 0.25)**, from −0.05 at 0 to +0.15 at 1, exactly **0 at 0.25**, the default. Above 0 the blacks lift to grey; below it the shadows crush.

White Clip — the ceiling on video: **1 + 0.3 (v − 0.5)**, from 0.85 at 0 to 1.15 at 1, exactly **1 at 0.5**, the default. Below 1 the picture is clipped at less than white; above 1 the clip is off the top of the range and nothing is clipped by this stage. The floor at 0 is always applied.

The White group

R Gain, B Gain — the red and blue gains about green, in linear light: **±6 dB**, 12 (v − 0.5) dB, exactly **0 dB at 0.5**. Green is the reference, as on a CCU. Because these come **before the knee**, a warm white (R Gain up) is compressed in red by the knee where a gain after the knee would clip it, and the harness measures that order.

Drift — a slow random walk on the R and B gains, in mireds: an Ornstein–Uhlenbeck walk with a 20-second time constant, **60 × v mireds RMS**. Default **0.15, which is 9 mireds RMS**: visible over a minute, never a jump. A mired moves the R and B gains in opposite directions by about 0.4%. The walk runs on the host's clock, so it drifts at the speed the show plays at, and it's seeded so a slider move never jumps the state. 0 is a camera that's warmed up. In the OpenFX build the walk is a function of the frame number instead; see OpenFX.

The Matrix group

Matrix — a 3×3 in linear light, applied after white balance. The presets are derivations with generic names, not any manufacturer's:

- **Identity**: nothing.
- **Standard** (the default): SMPTE 170M primaries to BT.709 primaries, derived from the chromaticities: a small red-and-green correction that most cameras carry as their base matrix.
- **High Saturation**: a saturation of 1.35 about BT.709 luma.
- **Film-like**: a saturation of 0.82 with a 4° hue rotation about the grey axis.

The luma of the matrixed picture is what the detail and skin stages work on.

Saturation — **2 v** about BT.709 luma, from 0 (monochrome) to 2, exactly **1 at 0.5**, the default. Multiplies whatever the preset did.

The Detail group

This is the aperture corrector, and the controls are the ones on a CCU's detail page.

Detail Level — the gain on the detail signal: **3 v**, from 0 (no detail at all) to 3. Default **0.3, which is 0.9**. The detail signal is the (−¼, ½, −¼) kernel on the luma at ±the delay, so **a step of height h overshoots by Level × h / 4** on one side and undershoots by the same on the other.

Crispening Freq — the delay, in pixels: **1 + 8 v**, from 1 pixel at 0 to 9 at 1, a whole number at every eighth of the slider. Default **0.125, which is 2 pixels**. This is the width of the halo: the overshoot lasts exactly the delay, whatever the edge's own width. A fractional delay lerps between two whole-pixel taps. The name is the CCU's, shortened to fit FFGL's limit of 16 characters.

H/V Ratio — horizontal against vertical detail: **H = min(1, 2 v), V = min(1, 2 (1 − v))**, so vertical only at 0, horizontal only at 1, **both at full at 0.5**, the default. The vertical detail uses the same delay in lines.

Coring — the edge height below which there is **no detail at all**: **0.25 v^2** in linear light, from 0 to 0.25, squared for resolution at the bottom. Default **0.3, which is an edge of 0.0225**. The dead zone on the detail signal is a quarter of that (a step of h peaks at h / 4). Below the coring, grain and fine texture are left exactly alone; above it, the overshoot is the full one less the dead zone.

Slider	edge height	what it looks like
0	0	every grain sharpened with the edges
0.3	0.0225	the default: the noise floor left alone, texture kept
0.6	0.09	only strong edges have detail; skin and fabric go smooth
1	0.25	plastic: only the hardest edges carry a halo

Level Dependence — how far the detail is reduced in the shadows: the detail gain is scaled by **1 – v (1 – min(1, Y / 0.2))**, so at 1 there is no detail at black and full detail from a luma of 0.2 up; at 0 the shadows are sharpened like everything else. Default **0.5**. Cameras have this because shadow noise is where detail hurts most.

Skin Detail — the detail gain inside the skin window is **1 – v**. Default **0.4**, so a face gets 60% of the detail the rest of the picture gets. At 1 there is no detail on skin at all; at 0 the window does nothing.

Skin Hue — the centre of the window, **360 v degrees** of hue on the linear colour, 0 at red and 60 at yellow. Default **0.0556, which is 20°**: skin. The two ends of the slider are the same hue.

Skin Width — the window's half-width, **5 + 55 v degrees**, from 5° to 60°. Default **0.2727, which is 20°**. The window is flat inside three-quarters of its width and smoothstepped to nothing at the edge, and it's gated on chroma, so a neutral is never skin, however its hue reads.

The Knee group

Knee On — on by default. Off, the picture above white clips; on, it compresses.

Knee Point — where the knee starts, **0.4 + 0.6 v in linear light**, from 0.4 to 1.0. Default **0.5, which is 0.7**. Below the point the chain is straight; above it the slope changes, continuous at the point.

Knee Slope — the slope above the point, **0.05 + 0.95 v**, from 0.05 to exactly 1 (no knee) at

1. Default **0.25, which is 0.29**: white at 1.0 linear lands at 0.787 linear, about 90% video, and that's why the defaults look milky at the top. The knee is **per channel**, in linear light, so a coloured highlight desaturates as it compresses. A per-channel knee is what most cameras call "knee".

With the point at 0.7 and the slope at 0.29, a linear value L above 0.7 becomes 0.7 + 0.29 (L − 0.7). At +6 dB (L = 2 for white) that's 1.08, still above the white clip, so the picture is a bright grey with the highlights flattened rather than a clipped one.

The Gamma group

Gamma — the opto-electronic transfer function's **exponent**, **0.35 + 0.2 v**, from 0.35 to 0.55, **BT.709's 0.45 at 0.5**, the default. The curve is a family: the linear segment's slope and the offset follow from the exponent by continuity at 0.018, and at 0.45 they're the standard's 1.099 and 4.5 to its printed precision. Lower is flatter with crushed highlights; higher is brighter and more contrasty. The clip was linearised through the same family at 0.45, so at the default the round trip is the identity.

Black Gamma — a lift below a quarter of video level, and nowhere else: a smooth bump on video of **0.056 × v** at its peak (a third of the way up to the level), zero at and above 0.25 video and with zero slope there. Default **0.2**, a lift of 0.011 at most. Lifts the shadows without touching the mid-tones or the highlights.

The Output group

Mix — the processed picture against the untouched clip, 0 to 1; **1 by default**. Zero is the clip as it arrived. The output carries the clip's own alpha throughout.

Show Detail — off by default. Instead of the picture, draws **the detail signal about mid grey**: what the aperture corrector is adding, after coring, level dependence and the skin window, before it's added. Flat areas are mid grey, edges are bright-and-dark pairs the width of the delay. Use it to set the delay and coring by eye, the way a camera engineer does on a monitor.

How it works

Once a frame, two passes on the GPU and a few constants in double on the CPU:

1. **Linear.** The clip is linearised (the inverse BT.709 curve, as a family at exponent 0.45), gained, white-balanced with the drift, and matrixed, into an RGBA32F buffer at picture size. The luma of the result goes into the buffer's alpha, because detail and skin work on it.
2. **Process.** Each output pixel reads its own linear value and eight neighbours (two per far tap, lerped by hand for a fractional delay), builds the detail signal, cores it, scales it by level and skin, adds it to all three channels, applies the knee, the OETF, black gamma, the pedestal and the white clip, and mixes with the source.

Every tap is a `texelFetch` at an integer coordinate, so nothing rests on the texture unit's filtering. The OETF's constants, the matrix and the drift's step are computed in double once a frame and handed over as floats. Nothing carries across frames except the drift, a double on the CPU stepped by the host's clock.

OpenFX — Resolve, Vegas, Nuke, Natron

The OpenFX build is the same chain. Everything between a slider and the picture's arithmetic (the control laws, the OETF's constants, the matrix, the drift's pull on the gains) is the same code as the Resolume build's; the two passes run on the CPU instead of the GPU, written out a second time statement for statement and checked against the GPU picture for picture (worst difference 1.6e-6, about 1/2,500 of an 8-bit step). What differs:

- **Drift follows the frame number.** An editing host renders frames in any order, alone, and on several threads at once, so the walk can't be carried from one frame to the next. It's replayed up to the frame being rendered (the same steps and the same seeded noise the Resolume build takes, each one frame of the clip's frame rate long), so a frame looks the same however and whenever it's rendered, and scrubbing shows the camera as it was at that frame. It starts from no drift at frame 0 and settles over the first minute, as the Resolume build does when the effect is added. The replay looks back at most 800 seconds; what it leaves out weighs less than the rounding of the number. Resolve's Fusion page reports the frame rate on the effect but not on its clips; Drift reads the effect's, and assumes 24 fps only where a host reports none.
- **Alpha.** The chain works on straight colour, as it does in Resolume. A premultiplied clip is divided by its alpha on the way in and multiplied back on the way out, which changes nothing on an opaque clip.
- **It runs on the CPU**: about 15 ms for a 1080p frame in a host lending it 8 threads of an Apple M4 Max (see Performance), against a fraction of a millisecond on the GPU in Resolume.
- **The detail delay is in pixels of the image the host renders.** A host rendering a half-resolution proxy gets halos twice as wide relative to the frame, as a camera of half the resolution would.

Nothing is left out: CCU has no audio input, no beat sync and no presets in either build, so every control carries over.

Performance

Measured by the offline harness on an M4 Max at the defaults, best of three runs of 60 frames after a warm-up, on a GPU that was busy with other things too:

	ms/frame	% of a 60 fps frame
1280×720	0.06	0.3%
1920×1080	0.17	1.0%
3840×2160	0.68–0.70	4.1%

It's two passes and eight texel fetches a pixel, so it's cheap. GPU memory is one RGBA32F buffer at picture size: 32 MB at 1080p, 127 MB at 4K. Nothing was timed inside Resolume, and nothing

was timed on Windows.

The OpenFX build renders on the CPU. On the same machine, both passes at the defaults:

	1 thread	8 threads	16 threads
1280×720	29 ms	4.5 ms	3.8 ms
1920×1080	66–71 ms	10 ms	8.5–8.9 ms
3840×2160	274 ms	39 ms	33 ms

The host decides how many threads it lends. In the fleet's test host, which lends 8, a whole 1080p render (converting the host's pixels in and out, replaying the drift and both passes) is about 15 ms, in 8-bit or float. Replaying the drift is about 0.5 ms of that at 60 fps (0.2 ms at 24), wherever on the timeline the frame is. Memory is two float copies of the frame: 66 MB at 1080p. I haven't timed it inside Resolve or any other commercial host.

If it looks wrong

Everything bright has gone milky grey. That's the knee at the defaults doing what a hot camera does. Raise Knee Point, raise Knee Slope toward 1, or turn Knee On off and let it clip.

The highlights clip hard and go white. Knee On is off, or Master Gain is high with the slope at

1. Turn the knee on.

The grain is sharpened. Coring is low. Raise it until the noise leaves the Show Detail view.

The picture is plastic. Coring is high, or Skin Detail is high with a wide window covering the whole picture's hue. Lower Coring; check Skin Hue and Skin Width.

A red or orange object has gone soft while the face is sharp. Skin Hue is on the wrong hue. Turn Show Detail on and move Skin Hue until the face's texture drops out and the object's comes back.

Warm highlights with white cores. A warm white balance (R Gain up) with the knee off: red reaches the ceiling first and the core of every highlight loses its warmth. Turn the knee on.

The colour slowly wanders. Drift is above 0. That's what it's for; set it to 0 for a warmed-up camera.

The blacks are grey. Master Black is above a quarter, or Black Gamma is high.

The picture is a grey field with bright-and-dark lines on it. Show Detail is on.

SW CCU isn't in the effects browser. Check the folder under Installing, and that Resolume was restarted.

The effect does nothing at all. A shader that won't compile looks just like that, and the actual message is in the log:

```
macOS    ~/Library/Logs/ccu/ccu.YYYY-MM-DD.log
Windows  %LOCALAPPDATA%\ccu\logs\ccu.YYYY-MM-DD.log
```

It records the build that was loaded, the GL vendor, renderer and version at load, which pass failed to compile if one did, and a buffer that could not be allocated.

Known limits

- **Never loaded into Resolume on macOS**, and nothing has driven the controls in a host there. I haven't tested how the twenty-three controls read in the inspector.
- **The clip stands in for scene light.** An 8-bit clip's white is the top of what the chain sees; only Master Gain and the R and B gains push anything above it, and the knee's character on footage comes from them.
- **The knee is per channel only.** There's no luma knee with chroma preserved.
- **The skin window is on the linear colour's hue**, not on a camera's post-gamma chroma vector.
- **The vertical detail uses the same delay as the horizontal**, in lines, lerped for a fraction; a real camera's line delays are whole lines.
- **The drift's mired-to-gain law is a judged constant** (0.4% per mired), not a Planckian locus, and the drift has never been watched over a minute in a host.
- **No audio input and no presets**, in either build.
- **The OpenFX build has rendered in one real host: DaVinci Resolve Studio 21.1 on macOS, as a Fusion tool**, where it matches the fleet's test host to within 1/255. Its first build failed every frame on Resolve's Fusion page, which reports no frame rate on its clips; the fixed build reads the effect's, which Fusion does report. Only the Fusion page has been tried, and it has never been run in Vegas, Nuke or Natron. Otherwise it's only run in the fleet's own OFX test host (8-bit and float, at frame 0 and at frame 300 of a drifting clip, alone and in sequence), where it matches the Resolume build byte for byte. It's never been handed a 16-bit clip or a proxy resolution, and I haven't tested what a real host's colour management does to the clip. Its Windows build has never been run, and its Linux build has only been loaded, not rendered.
- **Only ever run on an Apple M4 Max and on win-lab's software renderer**, although the macOS build contains an Intel slice. On Windows, see the note at the top of this guide.
- **There's a browser demo** at ccu-demo.stoatworks-labs.com. It runs the plugin's own shaders in WebGL2 with the CPU half ported to JavaScript, and the page lists what it doesn't reproduce.
- **Checked at up to 1280×720**, and only timed at 4K.

About

The last group, **About**, carries the plugin's name, version, licence and maker, and buttons that open this user guide (stoatworks-labs.com/software/ccu/guide/), the project page, the source on GitHub and the support page in your browser.

Reporting something

github.com/stoatworks-labs/ccu/issues. A screenshot, the Detail Level, Crispening Freq, Coring, Knee Point and Knee Slope settings, and the composition's resolution are usually enough. If the effect did nothing, attach the log.